AEGIS AUTONOMOUS SRE PLATFORM

Statement of Work & Software Subscription Schedule

Document	Statement of Work (SOW) & Subscription Schedule
Provider	Aegis Technologies (“Provider”)
Client	____________________________ (“Client”)
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Classification	Confidential

This document defines the scope, architecture, delivery roadmap, service levels, and commercial (license & subscription) terms under which the Provider will deliver and operate the Aegis Autonomous SRE Platform for the Client. It is a draft template; all figures are indicative and subject to Provider/Client legal and finance review before execution. Nothing herein constitutes legal or financial advice.

1. Purpose & Scope

Aegis is an autonomous Site Reliability Engineering (SRE) platform. It ingests production failure signals, runs a governed multi-agent reasoning loop to diagnose root cause, and proposes — and, where authorized, executes under human approval — a remediation, with a complete audit trail. This SOW covers (a) a subscription license to the Aegis platform, and (b) the implementation, integration, and capability-delivery services that bring the Client environment to production use.

In scope: deployment on the agreed tier (on-prem single-binary or horizontally-scaled cloud), integration with the Client's telemetry and version-control systems, the capability roadmap in Section 8, the service levels in Section 10, and the metered subscription in Sections 7 and 11. Out of scope items are listed in Section 13.

2. Vision & Positioning

The market value of SRE automation has shifted from advisory chatbots that summarize incidents to governed agents that securely interact with live production systems. Aegis is positioned on that shift along three axes:

- **Actionable, not advisory.** A safe, fail-closed, human-approved, fully audited action loop — the trust property enterprises require before an agent may touch production.
- **Open, standardized integration.** Live "eyes and hands" into Prometheus, Datadog, logs, and runbooks via the Model Context Protocol (MCP), rather than guessing from training data.
- **Hardware-adjacent depth.** A roadmap to diagnose and remediate the high-value, hardware-adjacent infrastructure (notably HPC / GPU clusters) that standard SaaS monitoring cannot comprehend — protecting the most expensive hardware in the Client estate.

3. Product Overview & Capabilities

Aegis decouples fast ingestion from slow reasoning so a failure storm queues rather than topples the system. The current platform performs autonomous code-level repair (crash → root-cause → validated patch → pull request under human approval); the roadmap generalizes this to a polymorphic *Signal* → *Diagnosis* → *Remediation* model where remediation is either a code change or a governed live action.

Capability	Description	Status
Telemetry ingestion	Kubernetes crash watcher, Sentry webhook, generic crash webhook; alert/metric adapters on roadmap	Live / extending
Autonomous diagnosis	Multi-agent loop: planner → researcher → executor → sandbox → reviewer	Live
Validated remediation	Patch applied to real source, compiled, optionally behaviorally tested; fail-closed	Live
Human-in-the-loop	Approval gate opens a real pull request / (roadmap) executes a governed action	Live
Observability	Prometheus metrics endpoint; distributed tracing on roadmap	Live / extending
Live action via MCP	Read metrics & execute approved runbooks against production	Roadmap (Stones 2–3)

Capability	Description	Status
HPC / GPU pack	GPU telemetry + GPU-specific remediation actions	Roadmap (Stone 5)

4. Architecture & Operating Processes

Two-tier, one codebase. Every infrastructure dependency (durable store, message broker, idempotency cache) sits behind an interface with two implementations, selected by a single profile variable. The on-prem tier runs as one zero-SaaS binary (SQLite + in-process queue + in-memory cache + local models); the cloud tier scales horizontally (Postgres + Redis Streams + Redis + separate worker pods).

Ingest process (fast path). Normalize signal → compute idempotency signature → atomic claim (de-duplicate) → persist as *pending* → publish to the broker → return in milliseconds (HTTP 202, or 429 back-pressure when the queue is full).

Process path (slow path). A worker consumes a delivery → runs the agent loop under a hard timeout (the “God-Node” kill switch) → writes a terminal status → acknowledges only after the status is durable (at-least-once delivery). On startup, events left *pending* by a prior crash are re-published (crash recovery).

Governance. Diagnosis never auto-ships: a security-review agent and a sandbox gate must both pass, and a human must approve before any change leaves the system. Every decision and action is recorded in the durable store for audit.

5. Core Algorithms

Algorithm	Mechanism & guarantee
Idempotent de-duplication	Atomic test-and-set claim on a content signature with TTL. Cloud: Redis SET NX EX (cluster-wide); on-prem: in-memory, TTL + LRU bounded. Closes the check-then-act race so identical signals are processed exactly once per window.
Crash signature	SHA-256 over *service_name : last-200-chars(crash_log)* — a stable key resilient to volatile leading log content.
At-least-once delivery	Acknowledge only after the terminal status is durably written; a worker crash mid-run leaves the message for redelivery.
Orphan recovery	Redis Streams XAUTOCLAIM reclaims deliveries idle in a crashed consumer’s pending list beyond a threshold, via a rolling cursor — making at-least-once real after a worker loss.
Sandbox patch application	Splice target→replacement (single, unambiguous occurrence) into a copy of the real source; reject if absent or ambiguous. The full patched file is validated, never a fragment.
Validation gate	Compile the patched file (argv, no shell); optionally run an operator-trusted reproduction command. Any failure, timeout, or unvalidatable case fails closed (never deploys).
Fail-closed safety	Infrastructure errors in review/execution default to “unsafe”; bounded retries with reviewer-feedback re-injection; a global per-incident timeout caps any run.
Rate limiting	Per-client sliding-window limiter with bounded-memory eviction of stale keys (abuse / cost protection).
Retrieval (RAG)	AST-structured code chunking + embedding-similarity retrieval over a codebase and an SRE-skills corpus; cache by content hash to avoid re-embedding unchanged inputs.
Policy & blast radius (roadmap)	Live actions default to dry-run; approval tier derived from tool risk and scope; post-action verification re-reads the triggering signal and auto-rolls back on regression.

6. System Variables & Parameters

All behavior is environment-driven; no secrets are hard-coded. The following are the primary tunable variables governing deployment, reliability, safety, and security.

Variable	Purpose	Default
AEGIS_PROFILE	Deployment tier: onprem | cloud	onprem
AEGIS_STORE / BROKER / CACHE	Override derived backends	(profile-derived)
AEGIS_DATABASE_URL / AEGIS_REDIS_URL	Cloud connection strings	—
AEGIS_DEDUP_TTL	De-duplication window (seconds)	300
AEGIS_QUEUE_MAX	In-process queue depth before back-pressure	1000
AEGIS_CLAIM_IDLE_MS	Redis orphan-reclaim idle threshold (ms)	60000
AEGIS_GLOBAL_TIMEOUT / AEGIS_MAX_RETRIES	God-Node timeout (s) / retry cap	120 / 3
AEGIS_WEBHOOK_TOKEN	Shared secret for webhooks/WS (required on cloud)	(unset)
AEGIS_SENTRY_SECRET	Sentry HMAC verification secret	(unset)
AEGIS_RATE_LIMIT_RPM	Per-client requests/minute (0 = off)	0
AEGIS_REPRO_COMMAND	Trusted behavioral reproduction command	(unset)
AEGIS_COMPILE_TIMEOUT / AEGIS_REPRO_TIMEOUT	Sandbox compile / repro caps (s)	30 / 120
SANDBOX_PROVIDER / VCS_PROVIDER	Sandbox (local|e2b) / VCS (github|gitlab|local)	auto / local
HERMES_MODEL / NEMOTRON_MODEL	Executor / reviewer model (local Ollama by default)	ollama/llama3

7. Billing & Metering Variables

The subscription is metered on the following billable variables. These are emitted by the platform's metrics subsystem and reconciled monthly. Unit prices are agreed in Section 11.

Billable variable	Definition	Billing basis
Protected services / nodes	Distinct services or hosts under active monitoring	Per unit / month
Incidents ingested	Signals accepted after de-duplication	Included to cap, then metered
Incidents auto-remediated	Incidents resolved to a verified terminal state	Per successful remediation (optional value-based)
Governed actions executed	Approved live actions executed via MCP (roadmap)	Per action / tiered
GPU nodes monitored (HPC pack)	GPU hosts under the HPC premium module	Per GPU-node / month (premium)
Environments & seats	Connected environments; approver/operator seats	Per environment / per seat
Data retention	Audit/incident history retention window	Per tier

Metering integrity: all billable counters are derived from the audited durable store and the metrics endpoint; the Client may reconcile against its own scrape of the same counters.

8. Delivery Plan & Capability Milestones

Capabilities are delivered in phases (“Stones”). Durations are indicative developer-effort estimates for a small team; each milestone has the acceptance criteria in Section 9 and is the trigger for the corresponding subscription tier to become active.

Phase	Deliverable	Eff. (dev-days)	Target
A — Foundation	Observability (metrics+tracing), eval harness & measured fix-rate, RAG wiring, HA state (Redis-backed approval/limit/fan-out)	25	~Wk 4
B — Generalize	Signal / Remediation model; validator supports patches and (dry-run) actions; zero regression	13	~Wk 6
C — MCP eyes	MCP read tools (Prometheus/logs); alert/metric ingestion adapters; live-context diagnosis	14	~Wk 9
D — MCP hands	Governed action execution: policy engine, gated execution, post-action verification, rollback, audit	19	~Wk 13
E — Productionize	Idempotent actions, multi-tenant isolation & quotas, CI eval gate, migrations, dead-letter queue, runbooks/SLOs	18	~Wk 17
F — HPC pack	GPU telemetry (DCGM/NVML/NCCL); HPC failure-mode knowledge; GPU remediation actions; HPC eval	32	~Wk 27

Critical path to the differentiated, sellable actionable agent is A→B→C→D (~Week 13). The HPC premium module (F) builds on the Phase-D action platform.

9. Acceptance Criteria & Testing

- **Foundation (A):** dashboards for ingest rate / queue depth / repair latency; the eval harness reports a baseline fix-rate; a 3-replica cloud run shows no duplicate processing.
- **Actionable (D):** on a staging environment, an alert is autonomously diagnosed, a remediation is human-approved, executed via MCP, and *verified* to resolve the signal; a forced failure auto-rolls back; zero unapproved mutations (audited).
- **Quality gate:** measured fix/remediation-rate meets the agreed threshold per domain (code, generic ops, HPC) before that capability is billed as production.
- **Each deliverable:** code + automated tests (or documented manual acceptance for infra-bound items) + metrics/audit coverage + no regression in the eval fix-rate.

10. Service Levels

Metric	Target (indicative)
Platform availability (cloud tier)	99.5% monthly (excl. planned maintenance)
Ingest acknowledgement latency (p95)	≤ 250 ms
Unapproved production mutations	0 (hard guarantee; all actions gated & audited)

Metric	Target (indicative)
Support response (Sev-1 / Sev-2)	1 business hour / 1 business day
Autonomy & remediation success	Reported monthly against the eval baseline

11. Commercial Model — License & Subscription

Aegis is delivered as a subscription license with metered usage. Figures below are **indicative placeholders** for negotiation and must be finalized by both parties' finance/legal teams; they do not constitute a binding quote.

Tier	Includes	Indicative basis
On-Prem License	Self-hosted zero-SaaS binary; local models; annual license + support	Annual license + support %
Cloud Standard	Hosted, scaled; code-repair + observability (Phases A–B)	Base / month + metered incidents
Cloud Actionable	Adds MCP live-context & governed actions (Phases C–D)	Higher base + per-action metering
HPC Premium Module	GPU telemetry & remediation (Phase F); add-on to any cloud tier	Per GPU-node / month
Implementation / Onboarding	Integration, configuration, eval calibration (one-time)	Fixed onboarding fee

Subscription billing is monthly or annual in advance; metered overages are reconciled in arrears against the Section-7 variables. Premium and action-based components activate only upon acceptance of the corresponding milestone (Section 9).

12. Security, Data Handling & Intellectual Property

- **Least privilege & auditability:** all webhook/WS access is authenticated; every action is gated, scoped, and recorded in the durable audit store.
- **Data residency:** the on-prem tier processes code and telemetry entirely within the Client boundary using local models (no third-party LLM egress) — suitable for regulated or air-gapped environments.
- **IP:** the Aegis platform, models, and tooling remain Provider IP; Client data, generated patches, and audit records remain Client property. License grants use for the subscription term only.

13. Assumptions, Dependencies & Exclusions

- Client provides timely access to target telemetry sources, version-control, and (for live actions) scoped credentials and approved runbooks.
- Remediation quality depends on the configured models; on-prem/local models trade some accuracy for privacy and cost — quantified by the eval harness.
- HPC and industrial-edge capabilities are roadmap items requiring a representative test environment and are pursued as scoped modules, not implied by the base subscription.
- Exclusions: bespoke integrations beyond the agreed connectors; on-call staffing of the Client environment; third-party software licenses.

14. Term, Renewal & Termination

Initial term: ____ months from the Effective Date, auto-renewing for successive equal periods unless either party gives ____ days' written notice. Either party may terminate for uncured material breach on ____ days' notice. On termination, the Client retains its data and audit records; license rights cease.

15. Acceptance & Signatures

By signing below, the parties agree to the scope, service levels, and commercial terms set out in this Statement of Work & Subscription Schedule, subject to the finalization of indicative figures noted herein.

For the Provider (Aegis Technologies)	For the Client
Signature: ____________________________	Signature: ____________________________
Name: ________________________________	Name: ________________________________
Title: _________________________________	Title: _________________________________
Date: _________________________________	Date: _________________________________

DISCLAIMER: This is a draft template prepared to support commercial discussion. All pricing, service levels, and durations are indicative and non-binding until finalized and executed by authorized signatories following Provider and Client legal and finance review. It does not constitute legal or financial advice.